

CAPSTONE PROJECT REPORT

1. Project Title:

Retail Bike Store Business Analysis.

2. Project Overview:

The project follows an end-to-end data analytics workflow, starting from raw data cleaning and preparation in **Microsoft Excel**, followed by data analysis using **SQL**, further cleaning and transformation using **Python and Pandas**, and finally creating an interactive **Power BI dashboard** for business reporting and visualization.

The objective is to convert raw business data into meaningful insights that can support data-driven decision-making.

3. Project Objectives

The main objectives of this project are:

- To clean and prepare raw sales data for analysis.
- To identify and handle missing, duplicate, and inconsistent data.
- To use Excel formulas for data transformation and validation.
- To perform business analysis using SQL queries.
- To clean and preprocess data using Python Pandas.
- To analyze sales, products, customers, stores, and inventory.
- To identify top-performing brands and products.
- To create an interactive Power BI dashboard.
- To present important KPIs and business insights visually.
- To support better business decision-making through data analysis.

4. Dataset Description

The dataset contains information related to bicycle sales and business operations.

The analysis covers different aspects of the business, including:

- Products
- Brands
- Categories
- Customers
- Orders
- Order items
- Stores
- Staff
- Sales

The dataset was initially reviewed and cleaned before being used for analysis.

5. Tools & Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning, formatting and formulas
SQL	Data extraction, transformation and analysis
Python	Data preprocessing and analysis
Pandas	Data cleaning and transformation
Power BI	Interactive dashboard and visualization

6. Project Workflow

The project was completed through the following workflow:

Raw Data → Excel Cleaning → Excel Formulas → SQL Analysis → Python/Pandas Cleaning → Power BI Data Modeling → Dashboard → Business Insights

This approach helped ensure that the data was properly prepared before creating the final dashboard.

7. Excel Data Cleaning

The first stage of the project was data cleaning using Microsoft Excel.

Activities performed:

- Removed duplicate records.
- Identified missing values.
- Checked inconsistent data.
- Corrected formatting issues.
- Standardized column values.
- Checked data types.
- Prepared data for further analysis.
- Used Excel formulas for calculations and validation.

Excel formulas used

Different Excel functions were used depending on the analysis requirements, such as:

- SUM
- AVERAGE
- COUNT
- COUNTA
- COUNTIF
- SUMIF
- SUMIFS
- IF
- Text and date functions

These formulas helped transform the raw data and calculate required business metrics.

8. SQL Data Analysis

SQL was used to extract and analyze information from the dataset.

The SQL analysis focused on:

- Total sales
- Number of orders
- Product performance
- Brand performance
- Category performance
- Customer analysis
- Store performance
- Revenue analysis
- Sales trends

SQL techniques used

The project involved SQL concepts such as:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- HAVING

- JOIN
 - INNER JOIN
 - LEFT JOIN
 - Aggregate functions
 - Subqueries
 - Conditional filtering
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9. Python & Pandas Data Cleaning

Python was used as another stage of data preprocessing and analysis.

The **Pandas** library was used to work with the datasets efficiently.

Activities performed using Pandas:

- Imported datasets.
- Inspected the data.
- Checked rows and columns.
- Identified missing values.
- Removed duplicate records.
- Checked data types.
- Converted columns into appropriate formats.
- Cleaned inconsistent values.
- Performed filtering and grouping.
- Prepared the cleaned dataset for visualization.

Pandas functions used

Some of the important functions included:

```
pd.read_csv()
df.head()
df.info()
df.shape
df.describe()
df.isnull().sum()
df.drop_duplicates()
df.fillna()
df.groupby()
df.sort_values()
df.dropna()
```

Python/Pandas helped automate and simplify several data-cleaning and transformation tasks.

10. Power BI Dashboard

After completing data preparation and analysis, the cleaned data was imported into Power BI.

An interactive dashboard was developed to provide a visual overview of business performance.

Steps Followed:

Created an interactive Power BI dashboard to analyze bicycle sales performance. The dashboard includes KPIs, sales and revenue trends, product and brand performance, category analysis, customer insights, store performance, inventory, and order status. Used charts, cards, slicers, and filters to provide an easy-to-understand view of key business insights.

The analysis produced several important insights:

1. **Trek was the top-selling brand** in the analyzed dataset.
2. **Trek Slash 8 generated the highest product revenue** in the analysis.
3. **Mountain Bikes recorded the highest total sales among the analyzed categories.**
4. Sales performance varies across products and categories.
5. Store-level analysis helps identify differences in business performance.
6. Inventory analysis provides visibility into product availability.
7. Order-status analysis helps monitor the overall order process.

8. Interactive Power BI filters make it easier to investigate specific business segments.

11. Conclusion:

This capstone project provided practical experience in performing an **end-to-end data analytics workflow**. The raw bicycle sales data was cleaned and prepared using Excel, analyzed using SQL, further processed using Python and Pandas, and finally transformed into an interactive Power BI dashboard.

The analysis identified important business insights, including the leading brand, top revenue-generating product, and highest-performing product category. The final dashboard provides an interactive platform for exploring sales, customers, products, stores, orders, and inventory.

Overall, the project demonstrates the ability to work with raw business data, perform data cleaning and analysis, create meaningful visualizations, and communicate business insights using modern data analytics tools.
